

CHE 201 HW 1
Due: 11.59 pm on 01/25/2020

1. (10 points)

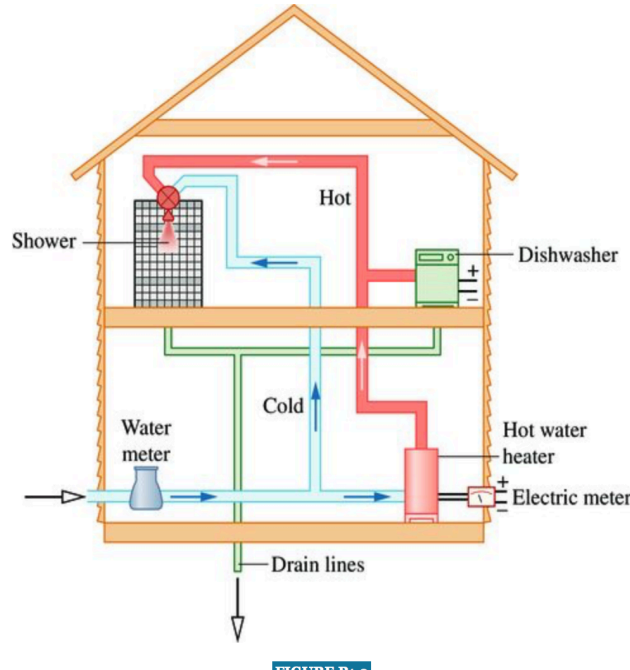
For problems 1.1–1.10, match the appropriate definition in the right column with each term in the left column.

- 1.1.  ___ Boundary
- 1.2.  ___ Closed system
- 1.3.  ___ Control volume
- 1.4.  ___ Extensive property
- 1.5.  ___ Intensive property
- 1.6.  ___ Process
- 1.7.  ___ Property
- 1.8.  ___ State
- 1.9.  ___ Surroundings
- 1.10.  ___ System

- A.** The condition of a system as described by its properties
- B.** A region of space through which mass may flow
- C.** Whatever is studied
- D.** A transformation from one state to another
- E.** A property whose value for an overall system is the sum of its values for the parts into which the system is divided
- F.** Everything external to the system
- G.** A fixed quantity of matter
- H.** A property whose value is independent of the size or extent of a system and may vary from place to place within the system at any moment
- I.** Distinguishes the system from its surroundings
- J.** A macroscopic characteristic of a system to which a numerical value can be assigned at a given time without knowledge of the previous behavior of the system

2. (20 points)

1.2 As illustrated in Fig. P1.2, water circulates through a piping system, servicing various household needs. Considering the water heater as a system, identify locations on the system boundary where the system interacts with its surroundings and describe significant occurrences within the system. Repeat for the dishwasher and for the shower. Present your findings in a memorandum.



3. (10 points)

Perform the following unit conversions:

- 122 in.³ to L
- 778.17 ft · lbf to kJ
- 100 hp to kW
- 1000 lb/h to kg/s
- 29.392 lbf/in.² to bar
- 2500 ft³/min to m³/s
- 75 mile/h to km/h
- 1 ton (=2000 lbf) to N

4. (10 points)

A closed system consists of 0.3 kmol of octane occupying a volume of 5 m³. Determine (a) the weight of the system, in N, and (b) the molar- and mass-based specific volumes, in m³/kmol and m³/kg respectively. Let $g = 9.81 \text{ m/s}^2$.

5. (10 points)

1.21 The pressure of the gas contained in the piston–cylinder assembly of Fig. 1.1 varies with its volume according to $p = A + (B/V)$, where A, B are constants. If pressure is in lbf/ft² and volume is in ft³, what are the units of A and B?

6. (10 points)

1.28 WP As shown in Figure P1.28, the exit of a gas compressor empties into a receiver tank, maintaining the tank contents at a pressure of 200 kPa. If the local atmospheric pressure is 1 bar, what is the reading of the Bourdon gage mounted on the tank wall in kPa? Is this a *vacuum* pressure or a *gage* pressure? Explain.

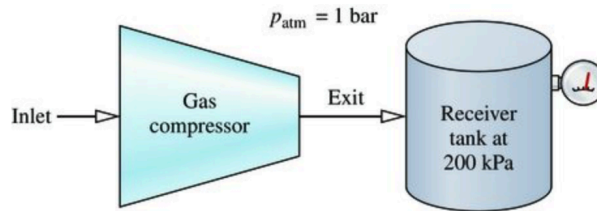


FIGURE P1.28

7. (10 points)

1.29 WP Liquid kerosene flows through a Venturi meter, as shown in Fig. P1.29. The pressure of the kerosene in the pipe supports columns of kerosene that differ in height by 12 cm. Determine the difference in pressure between points a and b, in kPa. Does the pressure increase or decrease as the kerosene flows from point a to point b as the pipe diameter decreases? The atmospheric pressure is 101 kPa, the specific volume of kerosene is $0.00122 \text{ m}^3/\text{kg}$, and the acceleration of gravity is $g = 9.81 \text{ m/s}^2$.

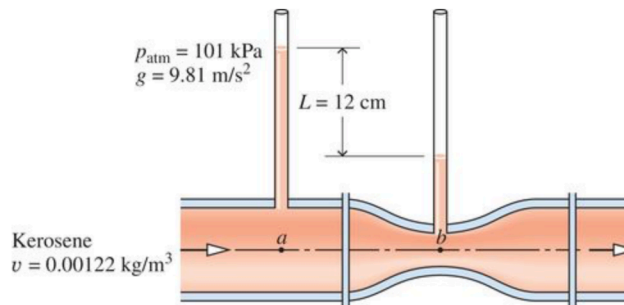


FIGURE P1.29

8. (20 points)

1.8D The *sphygmomanometer* commonly used to measure blood pressure is shown in Fig. P1.8D. During testing, the cuff is placed around the patient's arm and fully inflated by repeated squeezing of the inflation bulb. Then, as the cuff pressure is gradually reduced, arterial sounds known as *Korotkoff* sounds are monitored with a stethoscope. Using these sounds as cues, the *systolic* and *diastolic* pressures can be identified. These pressures are reported in terms of the mercury column length, in mmHg. Investigate the physical basis for the Korotkoff sounds, their role in identifying the systolic and diastolic pressures, and why these pressures are significant in medical practice. Write a report including at least three references.

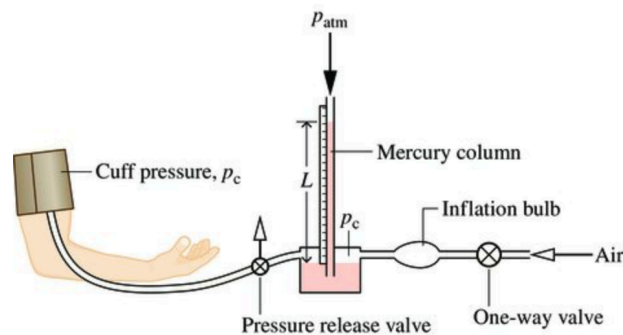


FIGURE P1.8D